



Fig. 7. The basic methodological paradigm and examples in *Interactive Exploration & Analysis*. (a) Compared with hyperbolic fisheye (left), the right one [205] reduce edge confusion. (b) Compared with original layout, distinguish a mixture of diverse edge bundling after user interaction [208]. (c) Optimize label position after users focusing [201].

feature extraction and computation by dictionary learning [193] and encoder-decoder structure [194]. On the other hand, mathematical optimization shows a wide application in 3D structure generation, many works utilize mathematical optimization to preserve shape compliance [180], compressive sensing in dynamic tomography [195] and volume reconstruction [196], even the medical lesion localization [197]. Firefly [177] optimizes the path of viewpoint change in rendering by using multi-objective optimization to balance rendered scene and path properties, as shown in Fig. 6(a). In field visualization, mathematical optimization could schedule parallel load [198], reconstruct the noise-filtered scalar field [199].

V. INTERACTIVE EXPLORATION & ANALYSIS

In this section, we conduct a comprehensive literature review on the interaction with mathematical optimization, since the interaction is a vital component in visualization after data transformation and representation. Interaction methods in visualization can be divided into *selection, reconfiguration, encode, filter, connect, abstract/elaborate, navigation, etc.* [200]. Based on the methods and objectives of interactions that integrate mathematical optimization, we summarize interaction methods into three categories as *Navigation, Filtering, and Reconfiguration*, which are introduced as follows. We also summarize the basic methodological paradigm of *Interactive Exploration & Analysis* in Fig. 7.

A. Navigation

Navigation methods provide an overview to help users access insights of visualization charts. During visual exploration, navigation often assists users in browsing visual elements or accessing the overall results of visualization charts initially. Then users conduct a deeper interactive exploration based on the navigation results. We introduce how to integrate mathematical optimization in navigation such as zoom, pan, and focus+context.

Zooming in or out in visualization charts is a widely-used interaction method to help users access insights into data. While in the zooming progress, the visual analysis burden such as deformation of graphs and overlaying of visual elements may

need to be solved or relaxed. In recent years, integrating mathematical optimization into the zooming interaction attracts the attention of researchers in many aspects such as laying out more labels in graphs [201], expanding routes in maps [202], and reducing the loss of context information on a small-screen [203].

Panning is to move visual elements from one to another position, which is helpful for users to compare neighbors or focus on one part of visualization charts. During the process of panning visual elements, the overlapping problem appears frequently, which results in researchers employing mathematical optimization models to relax the overlapping/collision of graphs. These works relayout treemap interactively after users panning [204].

Combining zooming and panning visualization charts can help users figure out overall patterns or access insights in sample charts. However, in complicated visualization charts, relying solely on pure zooming and panning proves inadequate for scenarios like large-scale graph visualization [205], because it fails to balance the point focusing and context accessing effectively. To relax this issue, researchers propose a combined interaction method named focus+context, such as fisheye. As the result shown in Fig. 7(a), Wang et al. [205] layout points in the fisheye diagram by quadratic optimization to preserve the structure of graphs. Researchers also focus on improving occlusion problems in traditional fisheye by multi-objective optimization during graph scaling fisheye [206]. Apart from the fisheye visualization, focus+context is also used in the metro map to relayout points and labels by a nonlinear optimization model after the users focuses [152] and deform the focus region by optimization to explore volume data more clearly in scientific visualization [207].

B. Filtering

Filtering helps users focus on their data of interest and mine the details of patterns by manipulating data with selecting, brushing, querying, etc. Filtering data in visualization is a well-established method to help users obtain further insights, particularly users who obtain an overview and conduct further analysis.

Pure filtering data hardly requires mathematical optimization, while researchers need to face mathematical optimization requirements in many specific scenes such as accelerating the result computation [209] and reducing the search space [210]. For example, during the process of filtering and selecting data based on topic modeling, Kim et al. employ a hierarchical NMF model to accelerate the computation of topic information, which is solved by a rank-2 non-negative matrix factorization algorithm [209]. To reduce search space for a topic model, they conduct further work which proposes an NMF model to feed target topic information after data filtering [210] and solve it by a rank-2 algorithm.

Filtering interaction technique itself may not need a complicated model, while employing mathematical optimization to accelerate the data computation [209], reduce search space [210], and provide candidate projection [210] is an effective method when users need interactive data filtering.